

1. From Classical Planning to Real-World Planning

- Classical planning assumes a fully observable, deterministic, static and discrete environment.
- Real-world planning must deal with uncertainty, partial observability, nondeterministic actions, exogenous events and continuous change.
- Plans must be executed while sensing, monitoring and adapting to what actually happens.
- The agent often interleaves planning, execution and replanning.

Key Idea: Real-world planning $\neq$ just finding a fixed sequence of actions; it requires continual interaction with the environment.

4. Uncertainty and Partial Observability

- The agent does not always know the exact current state of the world.
- It maintains a **belief state**: a set of possible states consistent with its observations.
- Sensing actions reduce uncertainty but may be noisy or incomplete.
- Planning must reason about what might be true, not just what is known for sure.

Belief state: a probability distribution or set of possible world states.

Example: Robot localization



7. Execution Monitoring and Plan Failure

- While executing a plan, the agent monitors the execution and compares observations with expected outcomes.
- A plan may fail because:
 - an action has an unexpected outcome.
 - preconditions are not satisfied.
 - the environment changes (exogenous event).
 - sensing reveals the goal is no longer reachable.
- Early detection of failure allows faster recovery.

Example: The robot tries to move through a door, but it is closed. The plan must be revised.

2. Why Real-World Planning is Harder?

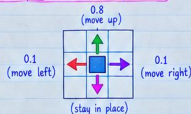
- Unknown initial state (partial observability).
- Actions may have multiple possible outcomes (nondeterministic).
- The environment may change on its own (exogenous events).
- Sensing is costly, noisy and sometimes inaccurate.
- Time matters: actions have duration and effects can overlap.
- Plans can fail, so the agent must monitor and adapt.
- Often we care about achieving and maintaining goals over time, not just a final state.

Real-world planning requires reasoning and decision making under uncertainty.

5. Nondeterministic Action Outcomes

- An action can have multiple possible results with some probability.
- Effects may depend on the current situation.
- Planning must consider all possible outcomes and choose actions that work well in these contingencies.

Example: Move action in a grid world



Nondeterministic outcomes require plans that are conditional on what actually happens.

8. Plan Repair vs. Replanning

- Local plan repair: make a small change to the existing plan to recover from a problem.
- Useful when the failure is minor and the rest of the plan is still applicable.
- Replanning: generate a new plan from the current (updated) state when repair is not enough.
- Often we combine both: try repair first, then replan if necessary.



3. Classical vs. Real-World Planning

Aspect	Classical Planning	Real-World Planning
Observability	Fully observable	Partially observable
Action outcomes	Deterministic	Nondeterministic
Environment	Static	Dynamic (exogenous events)
Time	Usually ignored	Actions take time and can overlap
Sensing	Not needed	Required (costly, noisy, limited)
Plan execution	Plan then execute	Interleave planning, acting and monitoring
Goal nature	Reach a goal state	Achieve and maintain goals

6. Sensing and Acting Loop

- In the real world, an agent repeatedly senses, plans (or updates its plan), acts, and monitors the outcome.
- Planning and acting are interleaved in a continuous loop.



Key points:

- Use observations to update the belief state.
- Choose the next action based on current knowledge.
- Monitor for success, unexpected events or failure.
- Replan or repair the plan when needed.

9. Conditional and Contingent Behavior

- A real-world agent must have conditional plans that specify what to do in different situations.
- Contingent planning: a plan is a tree or graph that branches based on observations and action outcomes.
- Each observation leads to a different sequence of actions.

Example: If the door is open, go through; otherwise, open the door and then go through.

10. Key Takeaways (Page 1)

- Real-world planning deals with uncertainty, partial observability and nondeterminism.
- The agent interleaves sensing, planning, acting and monitoring.
- Plans can fail; we need plan repair or replanning.
- Conditional and contingent behavior is essential.
- Success depends on continuous interaction with a changing environment.

1. Partial Observability in the Real World

- The agent cannot directly observe the full state of the environment.
- Sensors provide incomplete or noisy information.
- The agent must act based on a belief about the current state.
- Common in robotics, navigation, and monitoring domains.

Key Idea: Partial observability means the agent must consider multiple possible states rather than a single known state.

Example: A robot knows its location only approximately due to sensor noise.

4. Conformant Planning

- Find a fixed sequence of actions that achieves the goal regardless of the initial state (within a set).
- Does not use conditional plans or sensing during execution.
- Often conservative because the plan must work for all possible states.

When to use:

- No sensing available during execution.
- Simple environments with small uncertainty.
- Goal can be achieved by the same actions from all possible states.

7. Information Gathering

- Sometimes the agent must take actions specifically to gain information.
- Useful when a small cost of sensing leads to a much better plan.
- Information gathering reduces the expected cost by removing uncertainty.
- Requires reasoning about the value of information.

Example: A robot can move to a higher location to get a better view before deciding which path to take.

Trade-off:

- Cost of sensing (time, energy, risk).
- Expected benefit of reduced uncertainty.
- Better final plan and higher chance of success.

2. Belief States (Recap and Use)

- A belief state is a set (or probability distribution) over possible world states.
- It summarizes what the agent knows based on past actions and observations.
- Planning is done in the belief space instead of the original state space.
- Belief states allow the same framework to handle uncertainty, sensing, and nondeterminism.

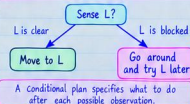
Belief state: set of all states consistent with the agent's observations (usually with probabilities).

Notation: b = belief state, s = world state
 $P(s)$ = probability of state s in b .

5. Conditional Plans

- A plan is a tree with branches based on observations (contingencies).
- The agent selects actions based on what it observes during execution.
- More flexible and often more efficient than conformant plans.

Representation: Use a plan tree with sensing actions and conditional branches.



8. Updating Beliefs after Observations

- Start with prior belief b (before observation).
- Take a sensing action and receive observation o .
- Update the belief using the observation model (Bayes' rule).
- New belief b' gives probabilities for states consistent with o .

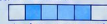
Belief update (intuitive):

$$P(s|o) \propto P(o|s)P(s)$$

(likelihood) (prior belief)
(observation model)

Example: Robot localization in a corridor

Before observation:



□ possible state
(equal probability)
■ most probable

After sensing "wall ahead":



$b' = \{0, 0, 0, 1.0\}$

3. Sensing Actions

- Sensing actions provide information about the current state.
- They do not directly change the world (usually), but change the agent's belief.
- Observations can be noisy or partial.
- Sensing actions reduce uncertainty and help choose better plans.

Example: A robot with a camera can sense whether a location is free or blocked (with some noise).

Types of sensing:

- Deterministic sensing (always correct).
- Noisy sensing (probabilistic observation).
- Partial sensing (observes only part of state).

6. Contingency Planning

- Plans are conditional on uncertain outcomes of actions and observations.
- Represented as a tree or graph.
- Each branch corresponds to a possible observation or outcome.
- Continue executing, sensing, and replanning as needed.

Features:

- Handles sensing and nondeterministic actions.
- Different sequences for different observations.
- Often interleave sensing and acting.
- Can be combined with online replanning if execution deviates from the plan.

9. Conformant vs. Conditional Planning

Aspect	Conformant Planning	Conditional Planning
Plan form	Single fixed sequence	Tree with branches
Uses sensing during execution	No	Yes
Handles uncertainty	Must work for all possible states	Adapts based on observations
Efficiency	Often conservative (longer plans)	Usually more efficient
Complexity	Simpler to compute	More complex
When to use	No sensing or simple problems	Sensing available and useful

10. Key Takeaways (Page 2)

- Partial observability requires belief states.
- Sensing actions provide information and reduce uncertainty.
- Conformant planning finds a fixed plan that works for all possible states.
- Conditional plans and contingency planning use observations to choose actions.
- Information gathering can lead to better overall performance.
- Beliefs are updated after each observation.
- Choosing when to sense is a trade-off between information cost and expected benefit.
- These ideas are essential for planning and acting in real-world environments.

"Better information today leads to better actions tomorrow."

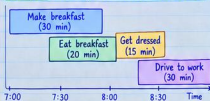
— AIMA (Ch. 12)

1. Temporal Planning

- Actions take time, so plans must consider temporal constraints.
- Each action has a duration, with start and end times.
- Temporal planning finds an ordering and timing of actions to achieve goals.
- Can allow actions to overlap when possible.
- Often represented with a temporal planning graph or as a scheduling problem.

Key Idea: A temporal plan specifies not just the order of actions, but also when each action starts and ends.

Example: Morning routine (goal: leave house)



4. Scheduling

- Scheduling assigns start times to actions subject to temporal constraints and resource limits.
- Goal is to find a feasible (or optimal) schedule that achieves the goals.
- Can optimize for makespan (finish time) or other criteria (e.g., cost, resource use).

Example: Simple schedule

Action	Start	End	Duration
Make breakfast	7:00	7:30	30
Eat breakfast	7:30	7:50	20
Get dressed	7:50	8:05	15
Drive to work	8:05	8:35	30

The schedule respects ordering constraints and resource availability.

7. Multiagent Planning

- Multiple agents act in the same environment, each with its own goals.
- Agents may need to coordinate, cooperate, or avoid conflicts.
- Planning can be centralized (one plan for all) or distributed (each agent plans individually).
- Must consider interactions such as resource sharing, collisions, or competing goals.

Example: Two robots moving boxes



Goals:

- R1: deliver Box A to location X.
- R2: deliver Box B to location Y.

"In the real world, successful action requires planning in time, handling resources, coordinating with others, and reacting to change." — AIMA (Ch.12)

2. Durative Actions

- Durative actions have a non-zero duration.
- Each action has preconditions at start, conditions that must hold during execution (invariants), and effects at the end.
- Important for modeling real-world actions like moving, cooking, or charging a battery.

Representation:

Action: Move(x, y)

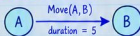
Duration: 5

Start preconditions: $At(x)$

Invariants: $\neg Blocked(x, y)$

End effects: $At(y), \neg At(x)$

Example: Move action



The agent is moving during the whole duration; it cannot also be at another location.

5. Resource Constraints

- Actions may require resources (e.g., a robot arm, a vehicle, or battery power).
- Resources have limited capacity, so actions that use the same resource cannot overlap if capacity is 1.
- Scheduling must ensure resource constraints are not violated.

Example: Two actions using a single robot arm



Since the robot arm has capacity 1, the actions cannot overlap.

8. Coordination

- Agents coordinate to achieve individual or joint goals.
- Techniques include negotiation, task allocation, and plan merging.
- Coordination is needed to avoid conflicts and to share resources efficiently.

Example: Coordination for a shared location

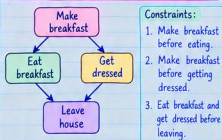


R1 and R2 must coordinate so that only one enters the hallway at a time.

3. Ordering Constraints

- Some actions must occur before others (precedence constraints).
- Constraints can be partial, allowing some actions to be unordered or parallel.
- Represented using a partial order (e.g., a directed graph) or constraint network.
- Enables flexibility and concurrency.

Example: Ordering constraints for daily tasks



Constraints:

- Make breakfast before eating.
- Make breakfast before getting dressed.
- Eat breakfast and get dressed before leaving.

6. Reusable vs. Consumable Resources

- Reusable resources: available again after the action completes (e.g., robot arm, vehicle, computer).
- Consumable resources: used up by an action and must be replenished (e.g., fuel, battery, supplies).
- Planning must track the amount of consumable resources over time.

Reusable Resource	Consumable Resource
Can be used multiple times	Used up and must be replenished
Typically has fixed capacity	Has a quantity that decreases
Available again after action ends	Not available until replenished
Examples: robot arm, vehicle, tool	Examples: fuel, battery power, materials

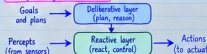
9. Reactive vs. Deliberative Control

Reactive Control	Deliberative Control
Responds directly to current percepts	Computes a plan using an internal model
Fast response time	Slower, more computation
No explicit planning	Explicit planning and reasoning
Good for dynamic and uncertain environments	Good for complex tasks that require long-term goals
Examples: reflex agents, subsumption architectures	Examples: classical planning, search-based planning

10. Hybrid Architectures

- Combine reactive and deliberative approaches.
- Deliberative component plans at a higher level.
- Reactive component handles low-level control and unexpected events.
- Common in real-world robots and autonomous systems.

Example: Layered architecture



1. Hierarchical Planning

- Solve complex problems by decomposing them into a hierarchy of smaller tasks.
- Each task is either a primitive action or a composite task that can be further decomposed.
- Reduces search effort by exploiting structure in the problem.
- Widely used in real-world planning systems.

Key Idea: Break a complex goal into a hierarchy of subgoals and tasks.

2. Hierarchical Task Networks (HTNs)

- A formalism for hierarchical planning.
- A task network is a set of tasks with ordering constraints.
- Tasks are decomposed using methods into subtasks until only primitive actions remain.
- More expressive and efficient than general-purpose planners for many real-world domains.

HTN Planner: Repeatedly decompose non-primitive tasks using methods to produce a valid plan (sequence of primitive actions).

3. Decomposition of Tasks (Example)

Example: Travel from home to a city.



Decomposition continues until all tasks are primitive actions that can be executed.

Key points:

- Use domain knowledge to decompose tasks.
- Methods specify how to achieve compound tasks.
- The final plan is a sequence of primitive actions.
- Hierarchical planning can greatly reduce the search space.

4. Reactive Systems

- Select actions based only on the current percept (no explicit planning).
- Fast and good for time-critical situations.
- Often use condition-action rules or finite-state controllers.
- Work well in dynamic and uncertain environments.

Reactive Agent



Key Idea: React immediately to the current situation using simple rules.

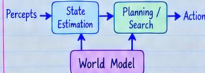
Example: Robot vacuum cleaner

If (obstacle ahead) → turn right
If (dirt detected) → clean

5. Deliberative Systems

- Explicitly represent the current state and reason about future actions.
- Use search or planning algorithms to compute a sequence of actions (a plan).
- Generally slower than reactive systems but can achieve better long-term outcomes.

Deliberative Agent



Key Idea: Build a model, plan ahead, and choose actions to achieve goals.

6. Hybrid Systems

- Combine reactive and deliberative approaches.
- Reactive layer handles time-critical responses and low-level control.
- Deliberative layer performs higher-level planning and goal reasoning.
- Common in real-world agents (e.g., robots, autonomous vehicles).

Hybrid Architecture



7. Continual Planning

- In dynamic environments, the agent continually monitors execution and updates the plan as needed.
- Planning and execution are interleaved.
- The agent revises its plan when the environment changes or new information is sensed.

Key Idea: Keep planning during execution, not just once at the start.

8. Anytime Planning

- Produce a valid plan even if computation is interrupted.
- The plan can be improved over time as more computation is available.
- Useful when time is limited but we still want a solution.
- Often based on best-first search, iterative deepening, or anytime variants of planners.

Key Idea: Return the best plan found so far and keep improving it as time allows.

9. Plan Repair

- Make a small change to the current plan when execution fails or the environment changes slightly.
- Reuse as much of the existing plan as possible.
- Faster than generating a completely new plan.
- Works well for minor disruptions.

Example: If a road is blocked, find an alternative route while keeping the rest of the plan unchanged.

10. Replanning

- Generate a new plan from the current state when the existing plan is no longer valid or too poor.
- Necessary when the environment has changed significantly or the failure is major.
- Can use classical planning, HTNs, or other planners to find a new plan.
- Often combined with plan repair: try repair first, then replan if needed.

Example: Robot's battery is low and cannot reach the goal → replan from the current state with battery constraint.

1. Chapter Summary

- Classical planning assumes a fully observable, deterministic, static and discrete environment.
- Real-world planning must handle uncertainty, partial observability, nondeterministic outcomes, exogenous events and continuous change.
- Plans and execution are interleaved; sense, plan, act and monitor.
- Contingent and conditional planning represent uncertainty and branching behavior.
- Plan repair and replanning allow recovery from unexpected situations.
- Execution monitoring detects failures early.
- Overall goal: build agents that continually achieve and maintain goals in dynamic, uncertain environments.

Key Idea: Real-world planning integrates sensing, reasoning, acting and monitoring to achieve robust, goal-directed behavior.

4. Plan Repair vs. Replanning

- Plan repair:** make a small change to the current plan to handle a failure or new information.
- Useful when the failure is minor and the rest of the plan is still applicable.
- Replanning:** generate a completely new plan from the current (updated) state.
- Useful when the plan is badly broken or conditions have changed significantly.

Plan repair is faster and reuses work, but replanning is more flexible when large changes are needed.

How to recover from a failure?



8. Practical Applications

Application	Planning Challenge	Real-World Issues
Robotics	Navigate and manipulate in dynamic environments	Sensing errors, moving obstacles, hardware failures
Autonomous vehicles	Route and behavior planning	Uncertain sensors, dynamic traffic, partial observability
Game playing	Long-term strategy under uncertainty	Opponent actions, unknown states
Service robots	Task planning in human environments	Human actions, exogenous events
Logistics and scheduling	Plan with changing constraints	Delays, resource availability, new tasks
Space exploration	Long-duration autonomous operation	Communication delays, unknown environments

2. Glossary of Key Terms

Term	Meaning
Classical planning	Planning in deterministic, fully observable, static and discrete environment.
Real-world planning	Planning under uncertainty, partial observability and continuous change.
Partial observability	Not all aspects of the state are known.
Nondeterministic action	Action may have multiple possible outcomes.
Exogenous event	Change in the environment outside the agent's control.
Conditional plan	Plan with branches based on observations or conditions.
Contingent plan	Plan represented as a tree/graph with conditional action outcomes.
Plan repair	Make a small change to fix a failed plan.
Replanning	Generate a new plan from the current state.
Conformant planning	Plan that achieves the goal without sensing (no observations).
Execution monitoring	Check progress, detect failures and update the plan if needed.

5. Conditional vs. Conformant Planning

- Conditional planning:** produce a plan with branches that specify what to do based on observations or conditions.
- Uses sensing during execution to choose the next action.
- Contingent planning:** similar idea, often represented explicitly as a tree or graph with nondeterministic action outcomes.
- Conformant planning:** produces a plan that achieves the goal without any sensing.
- Must succeed for all possible initial states consistent with the agent's knowledge.

Conditional planning uses sensing to choose actions; conformant planning does not sense and must work in all possible cases.

Types of Plans under Uncertainty



9. Final Chapter Takeaways

- Real-world planning deals with uncertainty, partial observability and nondeterminism.
- Planning and acting are interleaved in a continuous cycle with sensing and monitoring.
- Conditional, contingent and conformant planning provide ways to handle uncertainty.
- Plan repair and replanning allow recovery from failures and changes.
- The key to success is combining reasoning, sensing, acting and adaptation.

Real-world planning is more complex but also more realistic and useful than classical planning.

Execution monitoring is essential for robust real-world agents.

3. Classical vs. Real-World Planning

Aspect	Classical Planning	Real-World Planning
Observability	Fully observable	Partially observable
Determinism	Deterministic	Nondeterministic
Environment	Static	Dynamic (exogenous events)
Time	Usually ignored	Actions take time and can overlap
Sensing	Not needed	Required (costly, noisy, limited)
Plan execution	Plan then execute	Interleave planning, acting and monitoring
Goal nature	Reach a goal state	Achieve and maintain goals
Plan form	Linear sequence	Conditional/contingent policies (tree/graph)
Handling failure	Not considered	Plan repair or replanning

6. Common Exam Questions

- Compare classical and real-world planning. List and explain key differences.
- Explain why real-world planning is harder than classical planning.
- Describe conditional planning and contingent planning with examples.
- Explain plan repair vs. replanning. When would you use each?
- What is conformant planning? Give an example situation where it is needed.

7. Common Mistakes

- Assuming the environment is fully observable or deterministic.
- Ignoring the need for sensing and execution monitoring.
- Confusing plan repair with replanning.
- Thinking conditional and conformant planning are the same.
- Not considering exogenous events and changes in the environment.
- Assuming plans always succeed without failure handling.
- Forgetting that real-world planning and acting are interleaved processes.

"The real world is uncertain, dynamic and unpredictable — intelligent agents must plan, act, sense and adapt to achieve their goals."